Week 5 - Weekly Assignment covering C# Basics

All Exercises must be created Basic C# features only, No Arrays, Oops or advance options.

Exercise 1:

Mahendra Singh Dhoni has put up commendable achievements in his batting career with a boasting batting average of 51.08 (ODIs) & (38.09) in Tests and a striking rate of 89.27 (ODIs) & 59.11 (Tests). In a span of 10-11 years, not only has he become India's most successful captain, but he's also changed the limited-overs game in ways that one could only dream of.

If his entire Test batting statistics are referred, the runs scored by Dhoni in the continual matches followed a particular pattern of series. The series looks like 0, 6, 24, 60, 120, 210, 336, ... Predict the runs scored by Dhoni in this pattern of series upto "N" matches.

Exercise 2:

Mahi has learnt about the different properties of circles like radius, centre, diameter, circumference ... To make the tutorial class on circles more interesting, her teacher organises a game.

The players will be given 2 circles --- One circle A with radius ra and with central coordinate (xa, ya) and second circle B with radius rb and with central coordinate (xb, yb).

The player needs to say

- B is in A
- A is in B
- A and B intersect
- A and B do not intersect

You may assume that A and B are not identical. Can you please help Mahi in solving this game?

Exercise: 3

An organization wants to calculate an employee's Net Salary based on:

Basic Salary
HRA = 20% of Basic
DA = 10% of Basic
PF = 12% of Basic

If Basic Salary < 15,000 → No PF deduction.

Create a Class Library named SalaryCalculator and create a method in library as signature below:

public static double CalculateNetSalary(double basicSalary)

Consume this library in a console application. Get proper details of an individual Employee and display proper result. Implement Basic Exception Handling also in Library.

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Exercise 4:

Create an application to produce an Electricity bill using console application with proper format. Draw Boxes, lines and columns using characters only.

For input get CustomerId, CustomerName, Address, PhoneNumber, EmailId, Type of connection (Industrial, Business, Domestic, Agricultural), Previous Reading, Current Reading.

Calculate Unit Consumed and decide Electricity charges on slab basis as:

First 100 units → ₹1.5 / Unit consumed
Next 150 units → ₹2.5 / Unit consumed
Next 300 units → ₹4.5 / Unit consumed
Above 1000 Units → ₹7.5 / Unit consumed

Meter Rent based on Category should be charged as fixed amount.

Industrial → 2500
Business → 1500
Domestic → 1000
Agricultural → Free

Exercise 5:

Boxing is a martial art and combat sport in which two people throw punches at each other, usually with gloved hands. Historically, the goals have been to weaken and knock down the opponent. In December 2025, the World Boxing Council, World Boxing Association and the International Boxing Federation reached an agreement to standardize the names for January 2026 onwards, of the different weight categories under which Boxing matches are conducted.

Name of the weight category	Weight Range
light fly	Weight is less than or equal to 48 kg
fly	Weight is greater than 48 kg and less than or equal to 51 kg
bantam	Weight is greater than 51 kg and less than or equal to 54 kg
feather	Weight is greater than 54 kg and less than or equal to 57 kg
light	Weight is greater than 57 kg and less than or equal to 60 kg
light welter	Weight is greater than 60 kg and less than or equal to 64 kg
welter	Weight is greater than 64 kg and less than or equal to 69 kg
light middle	Weight is greater than 69 kg and less than or equal to 75 kg
middle	Weight is greater than 75 kg and less than or equal to 81 kg
light heavy	Weight is greater than 81 kg and less than or equal to 91 kg
heavy	Weight is greater than 91 kg

John is a good boxer and he wants to know under what category he should enrol his name.
Can you please help him out?

Input Format:
Input consists of an integer that corresponds to John's weight.

Output Format:
Output consists of a string that corresponds to any of the category names listed above or “Invalid Input”.
Print Invalid Input if the weight is negative or if the weight is greater than 120.

Enjoy Programming

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